

Nassau Candy Distributor Factory-to-Customer Shipping Route Efficiency Analysis

An Interactive Logistics Analytics and Business Intelligence Dashboard Using Python
and Streamlit

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Program / Context	Unified Mentor Data Analytics Internship Project
Project Type	Data Analytics, Logistics Intelligence and Interactive Dashboard
Primary Technologies	Python, Pandas, NumPy, Streamlit, Plotly, Folium, Matplotlib, Scikit-learn
Repository	github.com/chandolupraneethkumar05-oss/Nassau-Candy-Logistics-Analytics

Project Repository and implementation details were reviewed directly from the author's public GitHub repository. The dataset statistics reported in this paper were calculated from the repository's `cleaned_dataset.csv`.

Abstract

Logistics operations generate large volumes of transactional data, but raw shipment records alone do not provide managers with a practical view of route performance, factory productivity, regional demand, or profitability. This paper presents an interactive analytics solution developed for Nassau Candy Distributor to transform shipment-level data into operational and business intelligence. The system analyzes factory-to-customer routes, sales, gross profit, lead time, shipping modes, regions, product divisions, and factory performance through a multi-page Streamlit dashboard. The implementation uses Python and Pandas for data preparation and aggregation, Plotly for interactive analytical visualizations, Folium/Streamlit-Folium for geographic representation, and Streamlit for deployment of the decision-support interface. The repository's cleaned dataset contains 10,194 shipment records, 8,549 distinct orders, 5,044 customers, 59 states/provinces, 5 factories, and 15 products. The dashboard introduces a route efficiency measure defined in the implementation as gross profit divided by average lead time, enabling comparative ranking of routes. The study demonstrates how an accessible analytics application can connect operational metrics with management-oriented recommendations while retaining interactive filtering and downloadable reports. A data-quality limitation is also identified: the recorded order and ship dates produce unusually large lead-time values, so the resulting lead-time metrics should be interpreted as analysis of the supplied dataset rather than as validated real-world delivery durations.

Keywords: logistics analytics; supply chain; route efficiency; business intelligence; Streamlit; Python; data visualization; factory performance; lead time; decision support

1. Introduction

Modern distribution networks require organizations to balance delivery speed, profitability, shipping mode, production location, and customer geography. When these dimensions are analyzed separately, operational bottlenecks can remain hidden. A route can generate substantial sales but also exhibit poor delivery performance, while a factory may appear productive from a sales perspective but perform differently when lead time and route-level profitability are considered.

The Nassau Candy Distributor project addresses this problem by building a unified analytical dashboard around shipment-level data. The repository describes the goal as identifying opportunities to improve logistics efficiency through analysis of factory-to-customer routes, sales performance, lead time, factory productivity, and overall business performance. The resulting application is organized into Executive Dashboard, Sales Analytics, Route Analytics, Factory Analytics, and Business Insights modules.

The project therefore functions as a decision-support system rather than a static visualization report. Users can filter the analytical views, compare operational dimensions, inspect route performance, view factory locations, and download analytical reports. The application architecture uses Streamlit's page-navigation mechanism, with an entrypoint that defines the available pages and executes the selected page.

2. Problem Statement

Nassau Candy Distributor operates a multi-factory distribution network in which products move from factories to customers across different states and regions. The core analytical problem is to determine which factory-to-customer routes perform efficiently and which routes, factories, regions, or shipping modes may require managerial attention.

The project addresses the following questions:

- Which routes provide stronger profitability relative to their recorded lead time?
- Which factories contribute most strongly to sales and gross profit?
- How do shipping modes differ in volume, profitability, and recorded lead time?
- Which regions and product divisions contribute most to sales and profit?
- How can these findings be communicated through an interactive interface rather than isolated analytical scripts?

3. Objectives

The project was designed with the following objectives:

- Analyze factory-to-customer shipping routes and compare their performance.
- Calculate and rank route efficiency using the implemented efficiency metric.
- Monitor sales and gross-profit performance across regions, factories and products.
- Evaluate factory productivity and recorded lead-time behavior.
- Identify high-performing and low-performing operational segments.
- Provide executive-level business insights and recommendations.
- Deliver the analysis as an interactive, deployable Streamlit application.

4. Dataset and Data Preparation

The project uses the Nassau Candy Distributor dataset and stores both the source data and a cleaned version in the repository. The cleaned dataset contains 24 fields covering order identifiers, dates, shipping mode, customer information, geography, product information, sales, units, gross profit, cost, lead time, factory information, and derived factory-to-destination fields.

Characteristic	Value
Shipment records	10,194
Distinct orders	8,549
Distinct customers	5,044
States / provinces	59
Factories	5
Products	15
Total units	38,654
Total sales	\$141,783.63
Total gross profit	\$93,442.80

Table 1. Dataset profile calculated from the repository's cleaned dataset.

4.1 Data Cleaning and Transformation

The project documentation states that order and ship dates were converted to datetime format, invalid date records were removed, negative lead times were removed, and cleaned shipment records were prepared for analysis. The analytical application then loads the cleaned CSV and explicitly converts the date columns to datetime values before performing route and dashboard calculations.

4.2 Feature Engineering

Lead Time is represented as the difference between Ship Date and Order Date. Factory-to-state and factory-to-region route labels are present in the cleaned dataset, and the dashboard additionally creates a Route field by concatenating Factory and State/Province. Aggregations are then performed by route, factory, region, division, shipping mode, and product.

4.3 Data-Quality Observation

The supplied cleaned dataset has recorded lead times ranging from 904 to 1,642 days, with a mean of 1,320.84 days and a median of 1,274 days. These values are unusually large for conventional parcel-distribution interpretation and appear to result from the recorded date fields in the dataset. Consequently, this paper does not present these values as realistic delivery durations. They are retained as dataset-derived analytical measures, and future work should validate the source dates before using lead time for operational SLA decisions.

5. Methodology

The solution follows a data-to-decision workflow: source shipment records are cleaned and transformed; analytical features are derived; grouped metrics are calculated; interactive visualizations are generated; and the resulting evidence is exposed through a multi-page dashboard.

5.1 Analytical Metrics

Lead Time: $\text{Lead Time} = \text{Ship Date} - \text{Order Date}$. It measures the interval represented by the two recorded dates.

Route Efficiency Score: $\text{Efficiency Score} = \text{Route Gross Profit} \div \text{Route Average Lead Time}$. This is the metric implemented in the Route Analytics page to rank routes. A higher score indicates greater gross profit per unit of recorded average lead time within the dashboard's comparative framework.

Factory Performance: Factories are compared using aggregated sales, gross profit, shipment count, and average recorded lead time.

Regional Performance: Regions are compared using shipment count, sales, profit, and average recorded lead time.

5.2 Route Analytics Implementation

The Route Analytics module filters records by factory and region, counts unique factory-to-state routes, calculates total shipments and profit, and aggregates each route by shipment count, sales, profit, and average lead time. The resulting efficiency score is used to display the top and bottom routes, while a histogram visualizes lead-time distribution and a scatter plot compares factory sales, profit, and average lead time. The module also identifies the best route, the route requiring attention, and the fastest factory according to average recorded lead time.

5.3 Dashboard Architecture

Layer	Implementation
Data layer	Nassau Candy Distributor CSV + cleaned_dataset.csv
Processing layer	Python, Pandas, NumPy; datetime conversion and grouped aggregations
Analytics layer	Route, factory, region, division, product and shipping-mode metrics
Visualization layer	Plotly charts, tables, gauges, geographic maps
Application layer	Streamlit multipage dashboard with sidebar navigation and filters
Output layer	Interactive insights and downloadable CSV reports

Table 2. System architecture used by the implemented project.

6. Dashboard Design and Functional Modules

The dashboard is organized as a multi-page application. The repository's endpoint uses Streamlit's `st.Page` and `st.navigation` functions to group pages into Dashboard, Analytics, and Insights sections. This structure creates a clear separation between executive monitoring, analytical investigation, and management interpretation.

Module	Primary Purpose	Representative Outputs
Home	Project orientation and navigation	Project overview and access to analytical pages
Executive Dashboard	High-level operational overview	KPIs, sales, profit, lead time, trends, factory locations, route efficiency
Sales Analytics	Commercial performance	Sales, gross profit, trends, regions, factories, products, sales-vs-profit analysis
Route Analytics	Logistics performance	Route rankings, efficiency score, lead-time distribution, factory comparison
Factory Analytics	Factory-level evaluation	Sales, profit, lead time, locations, best-performing factory
Business Insights	Executive interpretation	Best region/factory, fastest factory, recommendations and management summary

Table 3. Functional decomposition of the Streamlit application.

6.1 Interactivity

Interactive filtering is central to the application. The Route Analytics page, for example, provides factory and region multiselect filters and recalculates route metrics on the filtered dataset. This allows users to move from an overall network view to a narrower operational segment without changing the underlying analysis code.

6.2 Visualization Strategy

The application uses bar charts for route ranking, histograms for lead-time distributions, scatter plots for factory sales-versus-profit comparison, KPI cards for rapid monitoring, tables for detailed route summaries, and gauges for the average route efficiency score. Plotly is particularly suitable for this implementation because it provides interactive charts that can be embedded in analytical applications.

6.3 Reporting

Several analytical modules provide CSV download functionality. This converts the dashboard from a read-only visualization into a lightweight reporting tool, allowing users to export route, sales, factory, or executive analytical outputs for further examination.

7. Results and Analytical Findings

The cleaned dataset contains 10,194 shipment records and generates total recorded sales of \$141,783.63 and gross profit of \$93,442.80. The dataset contains 5 factories, 59 states/provinces, and 15 products. The following results were calculated directly from the repository dataset.

7.1 Factory-Level Results

Factory	Shipments	Sales	Profit	Avg. Lead
Lot's O' Nuts	5,692	\$76,340.15	\$52,771.05	1,321.23
Wicked Choccy's	4,152	\$55,352.75	\$36,053.57	1,321.08
Secret Factory	217	\$8,587.50	\$4,344.70	1,321.87
The Other Factory	100	\$1,282.25	\$152.25	1,280.28
Sugar Shack	33	\$220.98	\$121.23	1,340.03

Table 4. Factory performance calculated from the cleaned dataset.

Lot's O' Nuts is the largest contributor in the supplied dataset by shipment count, sales, and gross profit. Wicked Choccy's is the second-largest contributor by these measures. The smaller factories contribute substantially less volume, so their route efficiency scores should be interpreted with awareness of lower sample sizes.

7.2 Regional Results

Region	Shipments	Sales	Profit	Avg. Lead
Pacific	3,253	\$46,301.53	\$30,485.94	1,322.19
Atlantic	2,986	\$41,197.24	\$26,973.70	1,322.75
Interior	2,335	\$32,037.60	\$21,282.49	1,323.09
Gulf	1,620	\$22,247.26	\$14,700.67	1,311.37

Table 5. Regional performance calculated from the cleaned dataset.

Pacific is the highest-sales region in the supplied data, followed by Atlantic and Interior. Gulf has the lowest total sales among the four regions but also the lowest average recorded lead time. This illustrates why logistics decisions should not rely on sales volume alone.

7.3 Shipping-Mode Results

Ship Mode	Shipments	Sales	Profit	Avg. Lead
Standard Class	6,120	\$85,490.35	\$56,424.46	1,314.33
Second Class	1,979	\$27,860.22	\$18,306.52	1,323.85
First Class	1,548	\$21,319.39	\$14,011.09	1,338.28
Same Day	547	\$7,113.67	\$4,700.73	1,333.44

Table 6. Shipping-mode performance calculated from the cleaned dataset.

8. Route Efficiency Analysis

The implemented route metric is calculated as gross profit divided by average lead time. This metric combines a financial outcome with the recorded time dimension and is useful for comparative ranking inside the dashboard. It should not be interpreted as a universal logistics KPI because it does not include distance, freight cost, service level, inventory cost, or customer priority.

8.1 Highest-Scoring Routes in the Supplied Data

Route	Shipments	Profit	Avg. Lead	Efficiency
Lot's O' Nuts → California	1,125	\$10,401.00	1,316.91	7.8980
Wicked Choccy's → California	823	\$7,421.32	1,320.33	5.6208
Lot's O' Nuts → New York	645	\$6,055.01	1,330.52	4.5509
Lot's O' Nuts → Texas	569	\$5,278.34	1,310.46	4.0279
Wicked Choccy's → New York	436	\$3,594.11	1,311.88	2.7397

Table 7. Top five routes by the implemented efficiency score.

The highest-ranked route in the supplied dataset is Lot's O' Nuts → California, followed by Wicked Choccy's → California and Lot's O' Nuts → New York. These rankings are driven by the interaction between route-level gross profit and average recorded lead time.

8.2 Routes Requiring Attention

Route	Shipments	Profit	Avg. Lead	Efficiency
The Other Factory → Maryland	1	\$0.25	1,274.00	0.0002
The Other Factory → Missouri	1	\$0.50	1,275.00	0.0004
The Other Factory → South Carolina	1	\$0.50	1,273.00	0.0004
Sugar Shack → Tennessee	1	\$0.60	1,272.00	0.0005
Sugar Shack → Washington	1	\$1.24	1,638.00	0.0008

Table 8. Five lowest-scoring routes by the implemented efficiency score.

The lowest-ranked routes in the supplied data are concentrated among lower-volume factories such as The Other Factory and Sugar Shack. Because these routes have relatively few shipments, ranking them solely by a ratio can be unstable. A production version should therefore combine the efficiency score with minimum-volume thresholds and confidence intervals before escalating a route as a confirmed operational bottleneck.

9. Discussion

The main contribution of the project is the integration of operational, financial, geographic, and executive perspectives in one application. Instead of requiring a manager to inspect several disconnected spreadsheets or notebooks, the dashboard provides a navigable analytical surface where the same shipment data can be examined from different decision perspectives.

The project also demonstrates the value of metric definition. The route efficiency score makes route comparison explicit, while factory and regional summaries provide context for interpreting the score. However, the metric is intentionally simple and should be considered a comparative analytics measure rather than a complete optimization objective.

9.1 Business Recommendations

- Prioritize high-volume routes with comparatively weak efficiency rather than focusing only on the absolute worst ratio.

- Monitor factory performance using sales, gross profit, volume, and validated delivery time together.
- Use regional analysis to distinguish demand concentration from logistics performance.
- Compare shipping modes using both service performance and profitability before changing mode policies.
- Introduce minimum shipment thresholds before classifying low-volume routes as operational bottlenecks.
- Validate the source date fields before using lead-time values for real-world service-level decisions.

10. Limitations

The current implementation is an analytics and decision-support dashboard, not a full logistics optimization engine. Its major limitations are:

- The recorded lead-time values are unusually large and require source-data validation.
- The efficiency score does not include physical distance, transportation cost, inventory cost, or service-level constraints.
- The current system is descriptive/diagnostic; it does not yet predict future delays or optimize routes automatically.
- Low-volume factories and routes can produce unstable ratio-based rankings.
- The project dataset represents a supplied analytical dataset rather than a live production shipment stream.

11. Future Scope

The repository identifies several natural extensions: machine-learning-based sales prediction, AI-assisted route optimization, demand forecasting, live shipment tracking, customer analytics, cloud deployment, and automated PDF report generation. These enhancements can evolve the project from descriptive business intelligence toward predictive and prescriptive logistics intelligence.

A stronger future architecture could add:

- Delay prediction using historical shipment attributes and validated target labels.
- Distance-aware route optimization using geographic coordinates and transportation constraints.
- Demand forecasting at product, region, and factory levels.
- Cost-versus-service optimization to recommend shipping modes.
- Real-time shipment ingestion and alerting for abnormal delays.
- Role-based dashboards for executives, logistics managers, factory managers, and analysts.
- Automated scheduled reports and cloud-based storage.

12. Conclusion

This research project demonstrates an end-to-end approach to logistics analytics using Python and Streamlit. By transforming shipment-level records into route, factory, regional, shipping-mode, sales, profit, and lead-time views, the system provides a practical foundation for data-driven logistics decision-making. The multi-page dashboard improves accessibility by allowing users to interactively filter data, compare performance, inspect routes, view geographic information, and download analytical reports.

The strongest aspect of the solution is its transition from raw data to an actionable management interface. At the same time, the project highlights an important principle of analytics: visualization cannot compensate for incorrect or ambiguous source data. Validating the recorded dates and expanding the efficiency metric would therefore be the highest-priority improvements before applying the system to real operational decision-making.

13. References

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Appendix A. Reproducibility

The project repository contains the application entrypoint, dashboard pages, data directory, requirements file, assets, charts, output directory, and project documentation. The application can be run locally with the Streamlit command specified in the repository documentation. The calculations in this paper were independently recomputed from the repository's `cleaned_dataset.csv` to avoid introducing unsupported numerical claims.